

Trial Higher School Certificate Examination

2008



Mathematics

Total Marks – 120

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Attempt ALL questions.
- Begin each question on a new booklet
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks)

Marks

- | | |
|--|---|
| a) If $a = 0.52$ find the value of $\frac{2+a^2}{2-a^2}$ to 3 significant figures. | 2 |
| b) Factorise $36m^2 - 9n^2$ completely. | 2 |
| c) Convert 54° to radians giving your answer in terms of π | 2 |
| d) Solve $ 3x - 2 \leq 10$ | 2 |
| e) Find the primitive of $3 - e^{-2x}$ | 2 |
| f) Simplify $x(2 - y) - y(3 - x)$ | 2 |

Question 2 - (12 marks)

Marks

a) The points $A(3, -3)$, $B(-3, -4)$ and $C(0, 3)$ are the vertices of a triangle ABC .

(i) Plot these points on the number plane.

1

(ii) Find the gradient of AC .

1

(iii) Find the angle of inclination of AC to the positive x -axis to the nearest degree.

1

(iv) Show that the equation of AC is $2x + y - 3 = 0$.

1

(v) Calculate the perpendicular distance of B from the side AC .

1

(vi) Hence find the area of $\triangle ABC$

2

(vii) Find the coordinates of D such that $ABCD$ is a parallelogram.

1

b) For what values of p will $x^2 + 5x + p$ be positive definite?

2

c) Find all the values of x between 0 and 2π for which $\sin x = -\frac{\sqrt{3}}{2}$

2

Question 3 - (12 marks)

Marks

a) Differentiate

(i) $\log_2(3x - 2)^2$

2

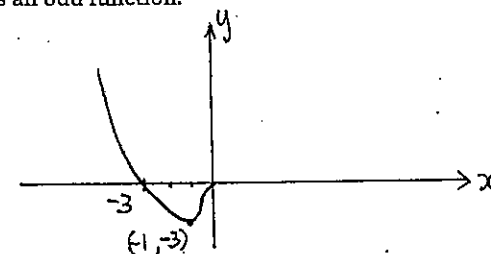
(ii) $x^2 e^x$

2

(iii) $\frac{1}{4x^4}$

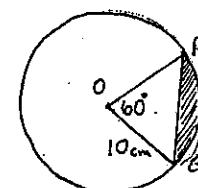
2

b) The following diagram shows the graph of $y = f(x)$ for $x \leq 0$. It is known that $f(x)$ is an odd function.



Copy the diagram onto your answer booklet and complete the graph for $x > 0$

2



From the diagram given of the circle centre O and radius 10 cm. Find the exact value of:

(i) the length of the minor arc QP .

2

(ii) the area of the minor segment cut-off by the chord QP .

2

Question 4 - (12 marks)

Marks

a) Find:

(i) $\int_4^9 x \sqrt{x} dx$

2

(ii) $\int \frac{3x}{x^2+1} dx$

2

b) Differentiate

(i) $\sin(5x + 3)$

2

(ii) $\log_2(\cos x)$

2

(iii) $e^x \tan 3x$

2

c) Find the value of

2

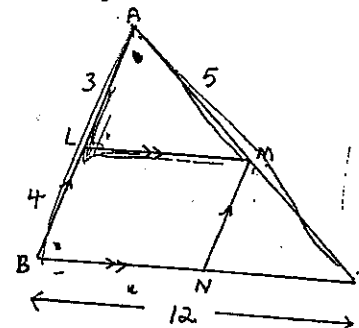
$$\sum_{n=1}^4 (3n^2 + 2)$$

Question 5 - (12 marks)

Marks

a) In the triangle ABC , L , M and N are on AB , AC and BC respectively so that LM is parallel to BC and MN parallel to AB

4



Find giving reasons:

(i) the length of MC .

(ii) the length of BN .

b) (i) Show that $\frac{\sec^2 x}{\tan x} = \frac{1}{\sin x \cos x}$

2

and hence

(ii) Find the value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin x \cos x} dx$

3

c) (i) Copy and complete the table of values for $y = \log_2(x + 2)$ in your booklet.

1

x	0	0.5	1	1.5	2
y	0.69	0.92	1.1	1.3	1.4

(ii) By using Simpson's rule with 5 function values, estimate the value of the integral $\int_0^2 \log_2(x + 2) dx$

2

Question 6 - (12 marks)

Marks

- a) The personal assistant to the CEO of Spendupbig Retail chain starts on an annual salary of \$30 000 with an annual increase of \$2 000.
- (i) Show that his salary forms an arithmetic sequence and write down a formula for determining his salary after n years with the company. 2
- (ii) Find his total earnings after he has worked for the company for 15 years. 2
- (iii) In which year will his salary be \$42 000? 1
- b) Find the value of the smallest term of the geometric series $4 + 10 + 25 + \dots$ that is greater than 10^{20} . Write your answer in scientific notation correct to 3 significant figures. 3
- c) (i) Find $\frac{d}{dx} \left(\frac{1}{2}x \sin 2x \right)$ 2
- (ii) Hence or otherwise find $\int x \cos 2x \, dx$ 2

Question 7 - (12 marks)

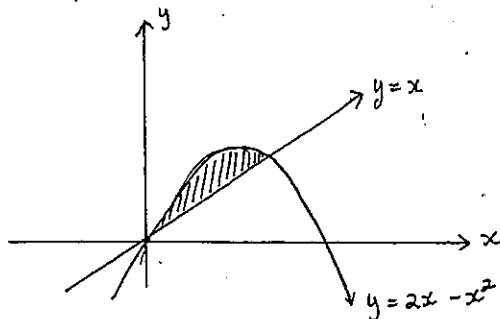
Marks

- a) (i) For what values of x does this geometric series have a limiting sum? 2
- $$2x + 6x^2 + 18x^3 + \dots$$
- (ii) Write an expression for the limiting sum and hence find the limiting sum if $x = \frac{1}{4}$ 2
- b) For the equation $y = 2 \cos 3x$ find:
- (i) the period. 1
- (ii) the amplitude. 1
- c) For the parabola $y = -3 - 4x - x^2$ find:
- (i) the vertex. 2
- (ii) the focal length. 1
- (iii) the focus. 1
- d) Sketch the region $x^2 + y^2 > 25$ 2

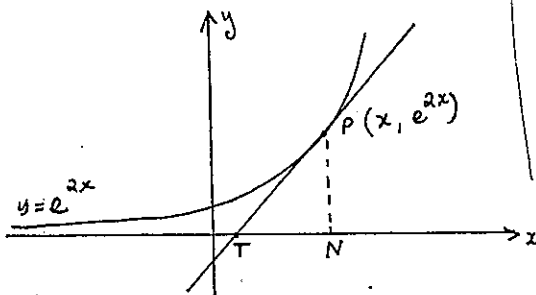
Question 9 - (12 marks)

Marks

- a) Find the volume of the solid formed when the region shaded in the diagram given is rotated around the x -axis. 4



b)



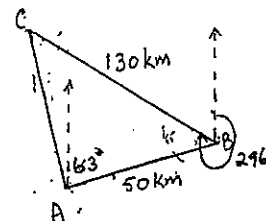
In the diagram above, $P(x_1, e^{2x_1})$ is a variable point on the curve $y = e^{2x}$. The tangent at P crosses the x -axis at T . The perpendicular from P to the x -axis meets the x -axis at N .

- Find the equation of the tangent to $y = e^{2x}$ at $P(x_1, e^{2x_1})$ 2
 - Find the coordinates of T . 1
 - Show that for all positions of P the length of TN is constant. 2
- c) Find the solutions of $2\sin^2\theta - \sin\theta = 0$ $0 \leq \theta \leq 2\pi$ 3

Question 8 - (12 marks)

Marks

- a) A ship sails 50km from Port A to Port B on a bearing of $063^\circ T$ then sails 130km from Port B to Port C on a bearing of $296^\circ T$ as shown in the diagram.



- Show that $\angle ABC = 53^\circ$ 2
- Find, correct to the nearest km, the distance of Port A to Port C . 2
- Find the bearing of Port A from Port C . 2

- b) The cost C (in dollars per hour) of running a boat depends on the speed v km/h of the boat according to the formula $C = 500 + 40v + 5v^2$

- (i) Show that the total cost of the trip of 100km is 2

$$T = \frac{50000}{v} + 4000 + 500v$$

- (ii) What speed will minimise the total cost of the trip. 4

Question 10 -- (12 marks)

Marks

a) For the function $y = x \log x$

(i) Write down the domain of the function.

1

$x > 0$

(ii) Write down the x -intercept of the function.

1

(iii) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

2

(iv) Hence find any stationary points and determine their nature.

2

(v) Given that $y \rightarrow 0^-$ as $x \rightarrow 0^+$ sketch the curve showing all relevant features.

2

b) At the beginning of each month Jane deposits \$600 into a bank account which pays 9% p.a. calculated monthly.

(i) How much will be in her account after four years.

2

(ii) Jane needed \$50 000. How much should she have deposited each month into this account for her to have reached her goal in the four years.

2

Question 1

$$2 + (0.52)^2 = 1.31267$$

$$2 - (0.52)^2 = 1.31 \text{ (3 sig. figs.)}$$

$$36m^2 - 9n^2 = 9(2m-n)(2m+n)$$

$$54^\circ = 54 \times \frac{\pi}{180} \text{ radians}$$

$$= \frac{3\pi}{10}$$

$$|3x-2| \leq 10$$

$$-10 \leq 3x-2 \leq 10$$

$$-8 \leq 3x \leq 12$$

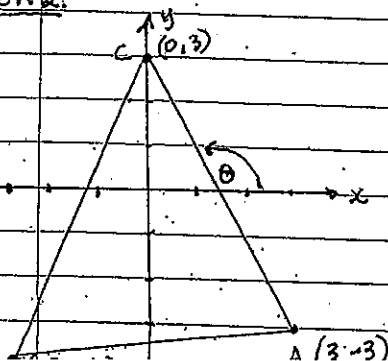
$$-\frac{8}{3} \leq x \leq 4$$

$$\int 3 - e^{-2x} dx = 3x + \frac{e^{-2x}}{2} + C$$

$$(2-y) - y(3-x) = 2x - xy - 3y + xy$$

$$= 2x - 3y$$

Question 2



$$(ii) m_{AC} = \frac{-3-3}{3-0}$$

$$= -2$$

$$(iii) \tan \theta = -2$$

$$\theta = 116^\circ 34'$$

$$\approx 117^\circ$$

$$(iv) y-3 = -2(x-0)$$

$$y-3 = -2x$$

$$2x+y-3=0$$

$$(v) d = \frac{|2x-3+1x-4-3|}{\sqrt{2^2+1^2}}$$

$$d = \frac{13}{\sqrt{5}}$$

$$(vi) AC = \sqrt{(3-0)^2 + (-3-3)^2}$$

$$= \sqrt{9+36}$$

$$= \sqrt{45}$$

$$= 3\sqrt{5}$$

$$\text{Area} = \frac{bh}{2}$$

$$= \frac{1 \times 3\sqrt{5} \times \frac{13}{\sqrt{5}}}{2}$$

$$= \frac{39}{2} \text{ square units}$$

$$(vii) D = (6, 4)$$

$$(b) b^2 - 4ac < 0, a > 0$$

$$x^2 + 5x + p \rightarrow a=1$$

$$\therefore a > 0$$

$$b^2 - 4ac < 0$$

$$25 - 4p < 0$$

$$-4p < -25$$

$$p > \frac{25}{4}$$

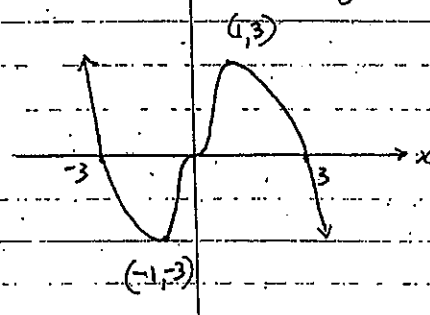
$$(iii) y = \frac{1}{4x^4}$$

$$y = \frac{1}{4} x^{-4}$$

$$y' = -\frac{4}{4} x^{-5}$$

$$= -\frac{1}{x^5}$$

(b) f(x) is odd. $\therefore$ rotational symmetry.



$$2) \sin x = -\frac{\sqrt{3}}{2}$$

$$\text{related angle} = \frac{\pi}{3}$$

$$x = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3}, \frac{5\pi}{3}$$

Question 3

$$(i) y = \log_e(3x-2)$$

$$y = 2 \log(3x-2)$$

$$y' = 2 \cdot \frac{1}{3x-2} \cdot 3$$

$$= \frac{6}{3x-2}$$

$$(ii) y = x^2 e^x$$

$$y' = 2x \cdot e^x + e^x \cdot x^2$$

$$= e^x(x^2 + 2x)$$

$$(c) i) l = r\theta \quad 60^\circ = \frac{\pi}{3} \text{ radians}$$

$$= 10 \times \frac{\pi}{3}$$

$$= \frac{10\pi}{3}$$

$$(ii) A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 10^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= 50 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

Question 4.

$$\begin{aligned} (1) \int_4^9 x \sqrt{x} dx &= \int_4^9 x^{\frac{3}{2}} dx \\ &= \left[\frac{2}{5} x^{\frac{5}{2}} \right]_4^9 \\ &= \frac{2}{5} (3^5 - 2^5) \\ &= \frac{2}{5} (243 - 32) \\ &= \frac{422}{5} \\ &= 84 \frac{2}{5} \end{aligned}$$

$$\begin{aligned} \int \frac{3x}{x^2+1} dx &= \frac{3}{2} \int \frac{2x}{x^2+1} dx \\ &= \frac{3}{2} \log_e(x^2+1) + C \end{aligned}$$

$$\begin{aligned} (1) y &= \sin(5x+3) \\ y &= 5 \cos(5x+3) \end{aligned}$$

$$\begin{aligned} (2) y &= \log_e(\cos x) \\ y &= \frac{1}{\cos x} \cdot -\sin x \\ &= -\frac{\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

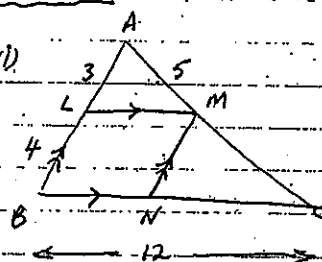
$$(iii) y = e^x \tan 3x$$

$$\begin{aligned} y' &= e^x \tan 3x + 3 \sec^2 3x \cdot e^x \\ &= e^x (\tan 3x + 3 \sec^2 3x) \end{aligned}$$

$$(c) \sum_{n=1}^4 (3n^2+2) = 5 + 14 + 29 + 50 = 98$$

Question 5.

(a) (i)



$$\begin{aligned} \frac{5}{MC} &= \frac{3}{4} \quad (\text{intercepts on } \parallel \text{ lines}) \\ MC &= \frac{20}{3} \end{aligned}$$

$$(ii) BN : NC = AM : MC \quad (\text{intercepts on } \parallel \text{ lines})$$

$$\begin{aligned} \text{Divide 12 in the ratio } 5 : \frac{20}{3} &= 3 : 4 \\ \therefore \frac{3}{7} \times 12 &= 5 \frac{1}{7} \end{aligned}$$

$$\therefore BN = 5 \frac{1}{7}$$

$$(b) (i) \frac{\sec^2 x}{\tan x} = \frac{1}{\sin x \cos x}$$

$$\begin{aligned} \text{LHS} &= \frac{\sec^2 x}{\tan x} \\ &= \frac{1}{\cos^2 x} \cdot \cot x \\ &= \frac{1}{\cos^2 x} \cdot \frac{\cos x}{\sin x} \\ &= \frac{1}{\sin x \cos x} \\ &= \text{RHS} \end{aligned}$$

(2)

$$\begin{aligned} \text{LHS} &= \frac{\sec^2 x}{\tan x} \\ &= \frac{1 + \tan^2 x}{\tan x} \\ &= \frac{1}{\tan x} + \tan x \\ &= \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \\ &= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} \\ &= \frac{1}{\sin x \cos x} \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} (11) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin x \cos x} dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan x} dx \\ &= \left[\log_e(\tan x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \end{aligned}$$

$$\begin{aligned} &= \log \left(\tan \frac{\pi}{3} \right) - \log \left(\tan \frac{\pi}{6} \right) \\ &= \log \sqrt{3} - \log \frac{1}{\sqrt{3}} \\ &= \log \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} \\ &= \log 3 \end{aligned}$$

(c)

x	0	0.5	1	1.5	2
y	ln 2	ln 2.5	ln 3	ln 3.5	ln 4
					1.39

$$\begin{aligned} A &\approx \frac{1}{3} \left[\ln 2 + \ln 4 + 4(\ln 2.5 + \ln 3.5) + 2 \right] \\ &\approx \frac{0.5}{3} [12.9] \\ &= 2.16 \end{aligned}$$

Q.6

$$A_1 = 30000 \quad A_2 = 32000 \quad A_3 = 34000$$

$$A_n = 30000 + (n-1)2000$$

$$= 28000 + 2000n$$

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$= \frac{15}{2} (2 \times 30000 + 14 \times 2000)$$

$$= \$660000$$

$$T_n = 42000$$

$$28000 + 2000n = 42000$$

$$2000n = 14000$$

$$n = 7$$

$\therefore$ after 7 years.

$$4 + 10 + 25$$

$$= 4 \quad r = 2.5$$

$$ar^{n-1} > 10^{20}$$

$$4 \times (2.5)^{n-1} > 10^{20}$$

$$(2.5)^{n-1} > \frac{10^{20}}{4}$$

$$(n-1) \log(2.5) > \log\left(\frac{10^{20}}{4}\right)$$

$$n-1 > \frac{\log\left(\frac{10^{20}}{4}\right)}{\log(2.5)}$$

$$n > \frac{\log\left(\frac{10^{20}}{4}\right)}{\log(2.5)} + 1$$

$$n > 49.7$$

the 50th term.

$$T_n = ar^{n-1}$$

$$T_{50} = 4 \times (2.5)^{49}$$

$$= 1.26 \times 10^{20}$$

(c) (i) let $y = \frac{1}{2}x \sin 2x$

$$y' = \frac{x}{2} \cdot \cos 2x \cdot 2 + \frac{1}{2} \sin 2x$$

$$= \frac{1}{2} \sin 2x + x \cos 2x$$

(ii) $\int \frac{1}{2} \sin 2x + x \cos 2x \, dx = \frac{1}{2}x \sin 2x$

$$\int \frac{1}{2} \sin 2x \, dx + \int x \cos 2x \, dx = \frac{1}{2}x \sin 2x$$

$$\int x \cos 2x \, dx = \frac{1}{2}x \sin 2x - \int \frac{1}{2} \sin 2x \, dx$$

$$\int x \cos 2x \, dx = \frac{1}{2}x \sin 2x - \frac{1}{2} \cdot \frac{-\cos 2x}{2}$$

$$= \frac{1}{2}x \sin 2x + \frac{1}{4} \cos 2x + C$$

Question 7

(a) (i) $2x + 6x^2 + 18x^3 + \dots$

$$r = 3x \quad |r| < 1$$

$$-1 < 3x < 1$$

$$-\frac{1}{3} < x < \frac{1}{3}$$

(ii) $x = \frac{1}{4}$

$$S_{\infty} = \frac{\frac{1}{4}}{1 - \frac{3}{4}}$$

$$= 2$$

b) (i) period = $\frac{2\pi}{3}$

(ii) amplitude = 2

2 $y = -3 - 4x - x^2$

$$x^2 + 4x = -y - 3$$

$$x^2 + 4x + 4 = -y - 3 + 4$$

$$(x+2)^2 = -y+1$$

$$(x+2)^2 = -(y-1)$$

i) vertex $(-2, 1)$

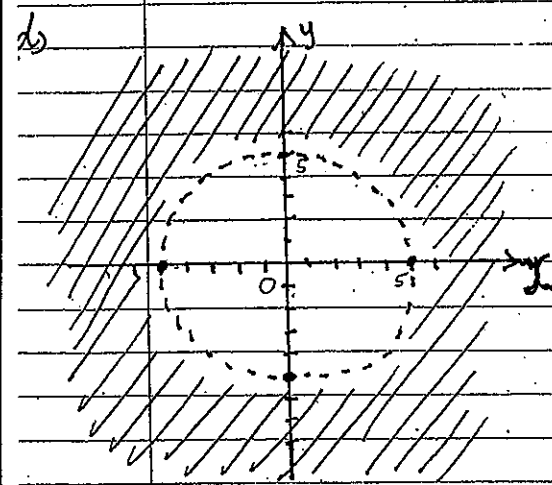
ii) $x^2 = 4ay$

$$4a = -1$$

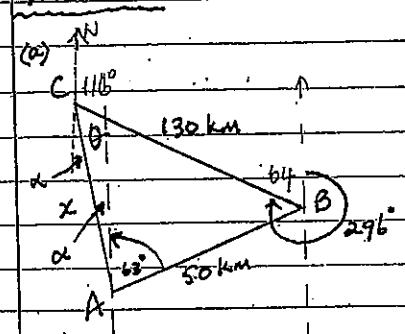
$$a = -\frac{1}{4}$$

$$\therefore \text{focal length} = \frac{1}{4}$$

iii) focus = $(-2, \frac{3}{4})$



Question 8



(i) Using alternate angles on $\parallel$ lines

$$\angle ABC = 116 - 63$$

$$= 53^\circ$$

(ii) $x^2 = 50^2 + 130^2 - 2 \times 50 \times 130 \times \cos 53^\circ$

$$x^2 = 11576.4$$

$$x = 107.59$$

$$= 108 \text{ km (nearest km)}$$

(iii) $\frac{\sin \theta}{50} = \frac{\sin 53}{108}$ ($\theta < 53^\circ$)

$$\sin \theta = \frac{\sin 53}{108} \times 50$$

$$\theta = \sin^{-1}\left(\frac{\sin 53}{108} \times 50\right)$$

$$= 21.6^\circ$$

$$\approx 22^\circ$$

$$\angle NCB = 116^\circ$$

$$\therefore \text{bearing} = 116^\circ + 22^\circ$$

$$= 138^\circ$$

$$\therefore \text{bearing of A from C} = 138^\circ$$

$$\text{time taken} = \frac{100}{v}$$

$$\text{total cost} = \text{cost/h} \times \text{time}$$

$$T = C \times \frac{100}{v}$$

$$= (500 + 40v + 5v^2) \times \frac{100}{v}$$

$$T = \frac{50000}{v} + 4000 + 500v$$

$$\frac{dT}{dv} = ?$$

$$= 50000v^{-1} + 4000 + 500v$$

$$= -50000v^{-2} + 500$$

$$= -\frac{50000}{v^2} + 500$$

$$= 0 \text{ for max or min}$$

$$0 = -\frac{50000}{v^2} + 500$$

$$\frac{50000}{v^2} = 500$$

$$50000 = 500v^2$$

$$100 = v^2$$

$$\pm 10 = v$$

$$= \frac{100000}{v^3}$$

$$\geq 10 \quad T'' > 0 \therefore \text{min.}$$

minimum cost when speed = 10 km/h.

Question 9.

$$(a) \quad y = x$$

$$y = 2x - x^2$$

$$x = 2x - x^2$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0, 1$$

$$\text{Vol} = \pi \int_0^1 (2x - x^2)^2 - x^2 dx$$

$$= \pi \int_0^1 4x^2 - 4x^3 + x^4 - x^2 dx$$

$$= \pi \int_0^1 3x^2 - 4x^3 + x^4 dx$$

$$= \pi \left[\frac{3x^3}{3} - \frac{4x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= \pi \left[x^3 - x^4 + \frac{x^5}{5} \right]_0^1$$

$$= \pi \left[(1 - 1 + \frac{1}{5}) - 0 \right]$$

$$= \frac{\pi}{5} \text{ cubic units.}$$

$$(b) (i) \quad y' = 2e^{2x_1} \quad P(x_1, e^{2x_1})$$

$$\therefore m = 2e^{2x_1}$$

$$y - y_1 = m(x - x_1)$$

$$y - e^{2x_1} = 2e^{2x_1}(x - x_1)$$

$$(ii) \text{ let } y = 0$$

$$\therefore -e^{2x_1} = 2e^{2x_1}(x - x_1)$$

$$-\frac{1}{2} = (x - x_1)$$

$$x = x_1 - \frac{1}{2}$$

$$T = (x_1 - \frac{1}{2}, 0)$$

$$T = (x_1 - \frac{1}{2}, 0)$$

$$N = (x_1, 0)$$

$$TN = x_1 - (x_1 - \frac{1}{2})$$

$$= \frac{1}{2}$$

TN is independent of x_1

$\therefore$ TN is a constant.

i.e. $TN = \frac{1}{2}$ for all x_1

(c)

$$2\sin^2\theta - \sin\theta = 0$$

$$\sin\theta(2\sin\theta - 1) = 0$$

$$\sin\theta = 0$$

$$2\sin\theta - 1 = 0$$

$$\theta = 0, \pi, 2\pi$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

Question 10.

$$(a) \quad y = x \log_e x$$

(i) domain $x > 0$

$$(ii) \quad x\text{-int. } y = 0$$

$$x \log_e x = 0$$

$$\log_e x = 0$$

$$x = e^0$$

$$= 1$$

$$\therefore (1, 0)$$

$$(iii) \quad y = x \log x$$

$$\frac{dy}{dx} = 1 + \log x + x \cdot \frac{1}{x}$$

$$= \log x + 1$$

$$\frac{d^2y}{dx^2} = \frac{1}{x}$$

(iv) $y' = 0$ for stationary pts.

$$\log x + 1 = 0$$

$$\log x = -1$$

$$x = e^{-1}$$

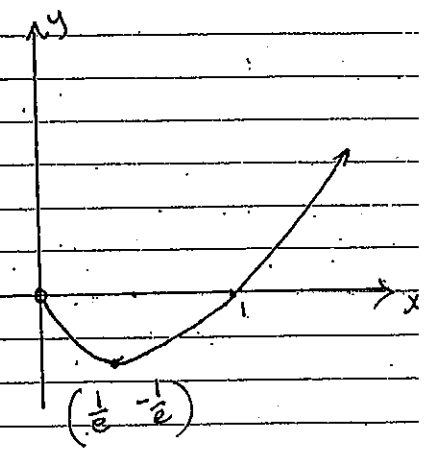
$$x = \frac{1}{e}$$

$$y'' = \frac{1}{x}$$

$$= \frac{1}{\frac{1}{e}}$$

$$= e > 0 \therefore \text{min.}$$

$$\left(\frac{1}{e}, -\frac{1}{e}\right)$$



$$\begin{aligned} \$600 \quad 9\% &= \frac{9}{12}\% \text{ per month} \\ &= 0.75\% \end{aligned}$$

$$1 = 600(1.0075)^{48}$$

$$2 = 600(1.0075)^{47}$$

$$3 = 600(1.0075)^{46}$$

$$8 = 600(1.0075)$$

$$= 600(1.0075) \quad 5 = 1.0075 \quad n = 48$$

$$8 = \frac{600(1.0075)(1.0075^{48} - 1)}{1.0075 - 1}$$

$$= \$34771.27$$

$$50000 = \frac{x(1.0075)(1.0075^{48} - 1)}{1.0075 - 1}$$

$$x = \frac{50000 \times 0.0075}{(1.0075)(1.0075^{48} - 1)}$$

$$= \$862.78$$